Managing Systemd Services and Targets

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Overview

Systemd Based Linux Systems:

- uniformity
- eco-system
- services
- targets
- logs and journalctl

System Uniformity

Systemd Eco-system

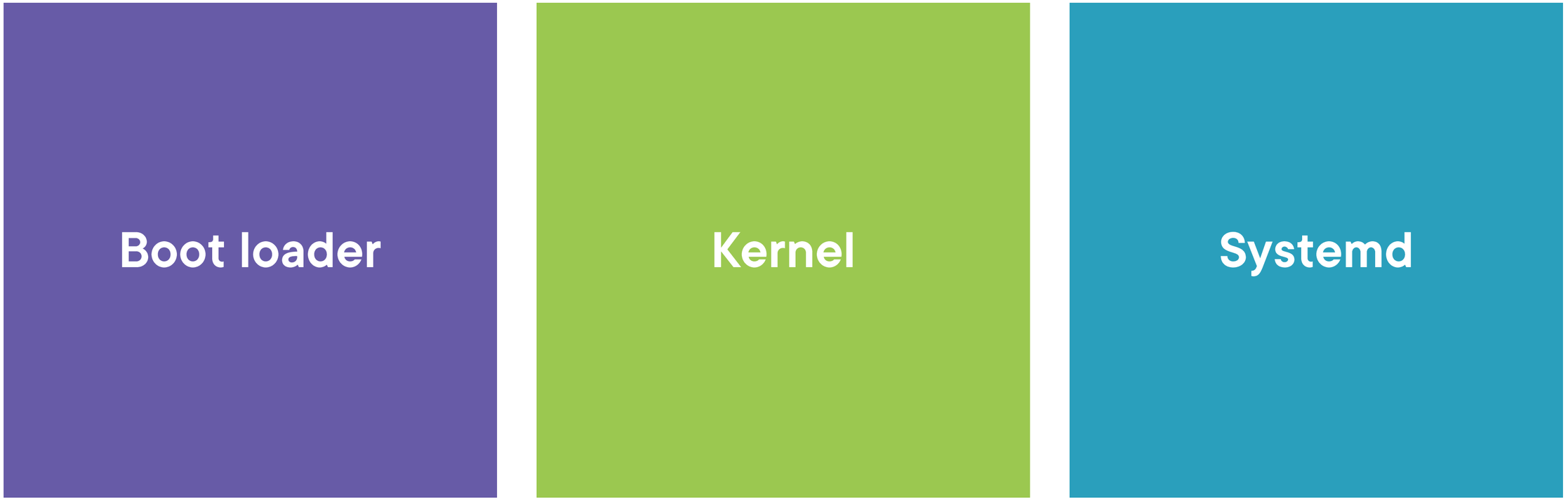
```
# hostnamectl set-hostname

# timedatectl set-timezone 'Europe/London'

# localectl set-locale LANG=en_GB.utf8
```

Demo

Linux Uniformity:

- hostnamectl
- timedatectl
- localectl

PID 1 = Systemd



Analyze Boot Time

```
# systemd-analyze

# systemd-analyzed blame
```

Demo

Analyze System Startup:

- process ID 1
- systemd-analyze

Using Systemctl to Manage Services

systemctl

```
# systemctl status cron
# systemctl enable --now cron
# systemctl disable --now cron
# systemctl cat cron
# systemctl edit cron --full
```

Demo

Managing Services:

- systemctl
- start/stop
- enable/disable
- --now
- mask
- status
- cat / edit

```
$ sudo -i
# fallocate -l 1G /root/disk1
# losetup /dev/loop10 /root/disk1
# parted /dev/loop10 mklabel msdos mkpart primary 0% 100%
# lsblk
```

Scenario

Creating our own disk file for use as storage in Ubuntu is easy. We use fallocate to create the file. Although the file is persistent it will need to be linked using losetup each time on startup and the partition table read with partprobe. Creating a service unit can automate the tasks for us.

The Problem with.....

The loop devices we have used do not persist

Creating a service unit we can script the device creation system start-up

Running both losetup and partprobe

losetup.service

```
[Unit]
Description=Setup Loop Device
DefaultDependencies=no
Before=local-fs.target
After=systemd-udevd.service

[Service]
Type=oneshot
ExecStart=/sbin/losetup /dev/loop1 /root/disk1
ExecStart=/sbin/partprobe /dev/loop1
RemainAfterExit=no

[Install]
WantedBy=local-fs.target
```

Demo

Working with Disk Files and Service Units:

- fallocate
- losetup
- parted
- systemctl edit

```
$ systemctl list-units --type target
$ runlevel
$ systemctl get-default
$ sudo systemctl  isolate multi-user
$ runlevel
$ sudo systemctl set-default multi-user
$ systemctl get-default
```

Targets

Targets group services together, a service is configured to be "installed" into a target if it is enabled for auto-start. Targets can include other targets allowing the grouping to become more global. For example, the graphical target will include the multi-user target and hence, all multi-user services will also start in the graphical target

Demo

Working with Targets

```
$ ls -l /var/log
$ cat /etc/rsyslog.conf
```

Traditional Logs

Traditionally, logs are stored in individual log files below /var/log. The main log file in Ubuntu is /var/log/syslog. The /etc/rsyslog.conf controls where log messages are sent to, along with the /etc/rsyslog.conf.d/ subdirectory.

```
$ sudo journalctl
$ sudo jounalctl --unit ssh
$ sudo journalctl --unit ssh --since today
```

Journal Log

The systemd journal log amalgamates logs centrally and can be filtered using journalctl

```
$ sudo grep -i 'storage' /etc/systemd/journald.conf
$ man 5 journald.conf
$ sudo sed -i 's/#Storage=auto/Storage=persistent/' /etc/systemd/journald.conf
$ sudo grep -i 'storage' /etc/systemd/journald.conf
$ sudo systemctl restart systemd-journald
```

Persisting Journal Logs

The journal log may only be resident in memory, we can configure persistent storage if required.

Demo

Working with Logs:

- viewing logs and the tail command
- viewing the journal log
- persisting the journal log

Summary

Managing Services

- systemd and uniformity
  - hostnamectl
  - localectl
  - timedatectl
- managing services using systemctl
- creating service units
- managing targets
- managing logs and journalctl

Up Next:
Scripting Automation in Linux